

Amendment of China's Biotechnology Laws in Relation to the Prevention and Containment of the COVID-19 Pandemic

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1. INTRODUCTION

AT THE END OF 2019, WUHAN, THE CAPITAL CITY of Hubei Province in China, reported the discovery of a novel coronavirus disease (COVID-19),¹ which spread rapidly and swept the world in early 2020. The outbreak of COVID-19 is considered an international public health emergency by the World Health Organization (WHO), as well as a major Chinese public health emergency by China's National Health Commission.² COVID-19 has become a pandemic worldwide and infected more than 29 million people, with more than 92,000 deaths as of September 14, 2020.³

To contain this sudden outbreak, China's central government insisted on principles of "people supremacy" and "life supremacy." Many measures have been taken simultaneously to fight COVID-19, drawing on the strengths of China's socialist institutions and legal system. From January 24, 2020 to March 8, 2020, a total of 346 medical teams composed of 42,600 medical workers and 965 public health workers from across the country and the armed forces were dispatched to Wuhan. Through May 31, a total of RMB 162.4 billion had been allocated by all levels of Chinese government to

fight the virus; relevant authorities had approved 19 applications for clinical trials of 17 medicines and vaccines for coronavirus prevention and control, and conditionally approved the applications for sale of two medicines.⁴ After months of work, COVID-19 has been contained in China, and all its provinces have lowered their states of emergency so that at present, first-level (more serious) response has been canceled across the country. Facing this global pandemic, the central government responded swiftly, containing the pandemic through its social organization capabilities in a short time. China's achievement in containing COVID-19 not only depended on medical technology and interventions, but also benefited from China's biotechnology legal system, including its public health legal system, emergency legal system, and animal protection legal system. These legal systems ensured that prevention and containment activities at all social

¹*Naming the Coronavirus Disease (COVID-19) and the Virus That Causes It*, WORLD HEALTH ORGANIZATION (Feb. 2020), [https://www.who.int/emergencies/diseases/novel-coronavirus-2019/technical-guidance/naming-the-coronavirus-disease-\(covid-2019\)-and-the-virus-that-causes-it](https://www.who.int/emergencies/diseases/novel-coronavirus-2019/technical-guidance/naming-the-coronavirus-disease-(covid-2019)-and-the-virus-that-causes-it)

²*WHO Coronavirus Disease (COVID-19) Dashboard*, WORLD HEALTH ORGANIZATION (Sept. 14, 2020), https://covid19.who.int/?gclid=Cj0KCQjwqfz6BRD8ARIsAIXQCf221a4idDJmUsLrJBanrWBPgabHP5xOfa-bo-akYINmQv79wyzcfGwaAq0NEALw_wcB

³*Coronavirus Disease (COVID-19) Situation Report—134*, WORLD HEALTH ORGANIZATION (June 2, 2020), https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200602-covid-19-sitrep-134.pdf?sfvrsn=cc95e5d5_2

⁴*Fighting Covid-19 China in Action*, STATE COUNCIL INFORMATION OFFICE OF CHINA (June 7, 2020), http://www.xinhuanet.com/english/2020-06/07/c_139120424.htm

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levels were carried out in an orderly and legal manner,⁵ providing legal response guidelines for cutting off the viral transmission route, organizing emergency relief and measures, and eradicating the virus from the biologic source.

2. CHINA'S APPROACH TO COVID-19 PREVENTION AND CONTAINMENT AND CURRENT CHINA'S BIOTECHNOLOGY LAWS

China has not only succeeded in its domestic anti-pandemic affairs, but also provided the international community with vital data about the virus, making a significant contribution to the prevention and containment of COVID-19 worldwide.

Firstly, China won time for global anti-pandemic measures. Wuhan is the first city in the world to have responded to the discovery of COVID-19. As a transportation hub in the middle of China, Wuhan faced tremendous pressure on the eve of the "Spring Migration." To control the movement of population and curb the spread of COVID-19, Wuhan's local government announced on January 23, 2020, that the city's public transportation would be suspended and the exits from the city closed. The shutdown measures in Wuhan immediately worked. On April 29, 2020, *Nature* published a paper showing that between January 22, 2020 and January 23, 2020, *i.e.*, immediately after Wuhan's lockdown, the efficacy of the quarantine was manifested in a reduction of 52% and 38%, respectively in inter- and intra-provincial population outflow, and a further reduction of 94% and 84%, respectively, on January 24, 2020 (as compared with January 23, 2020).⁶ An article from *Science* argues that "Wuhan Shutdown" was crucial and effectively prevented the large-scale spread of COVID-19.⁷ Inspired by the shutdown measures in Wuhan, other cities in China quickly followed and introduced their own public transportation control measures to restrict the movement of people. These measures gained time for the global implementation of prevention and containment measures against COVID-19.

Secondly, China has avoided a large-scale outbreak and spread of COVID-19. On May 4, 2020, a paper published in *Nature* concluded that the non-pharmaceutical containment strategies taken by the Chinese government were highly effective: traffic restrictions, isolating ill persons, contact tracing, quarantining exposed persons, and school and workplace closures, among other measures.⁸ This paper postulated that if interventions in China had

been conducted one week, two weeks, or three weeks later than they were, cases may have shown a three-fold, seven-fold, or even 18-fold increase, respectively.⁹ Moreover, it also estimated that COVID-19 cases would likely have shown a 67-fold increase without these non-pharmaceutical interventions and reached 7,659,775 cases by February 29, 2020.¹⁰ As a populous developing country, China minimized COVID-19 cases through rapid social mobilization, averted a broad February outbreak in China, East Asia, and even the world. The opportunity was thus created for other countries to study the virus, isolate strains, screen vaccines, and develop and implement prevention and containment measures.

Thirdly, China afforded experience for global pandemic prevention and containment. WHO has recommended four types of pandemic prevention and containment measures: large-scale testing, quarantining infected cases, contact tracing, and stay-at-home orders and the effectiveness and feasibility of these measures have been verified in China. At the governmental level, the Chinese government and the World Health Organization organized joint meetings to share the experience of pandemic prevention and containment.

Moreover, shutting down a city is only the first half of pandemic response; how to safely and effectively lift restrictions is also critical. China's experience has also provided guidance for other countries in reopening their cities. For example, the State Council issued the "Notice on Effectively Strengthening Scientific Prevention and Contain of the Pandemic and

⁵From the perspective of epidemiology, the most effective way to interrupt the transmission of COVID-19 is to cut off unnecessary and unsafe contact between humans, and the law is exactly the most effective way to control human behavior. Thus, it is to use the law to prevent and contain COVID-19. See Wang Chenguang, *Logic and Implementation of the Optimization of the Legal System for Pandemic Prevention and Contain*, 3 PEKING UNIVERSITY L. J. (June 2020).

⁶J.S. Jia, X. Lu, Y. Yuan, *et al.*, *Population Flow Drives Spatio-temporal Distribution of COVID-19 in China*, 582(7812) NATURE 389, 390 (June 18, 2020).

⁷Huaiyu Tian, Yonghong Liu, Yidan Li, *et al.*, *An Investigation of Transmission Control Measures During the First 50 Days of the Covid-19 Pandemic in China*, 368(6491) SCIENCE 638 (May 8, 2020).

⁸S. Lai, N.W. Ruktanonchai, L. Zhou, *et al.*, *Effect of Non-Pharmaceutical Interventions to Contain COVID-19 in China*, 585 NATURE 410 (May 4, 2020), <https://doi.org/10.1038/s41586-020-2293-x>

⁹*Id.*

¹⁰*Id.*

Resuming Work and Production of Enterprises in an Orderly Manner” on February 8, 2020, and required governments at all levels to gradually resume work. Although many other major countries are still in the first half of the response (implementing restrictive measures), there is a need to understand and formulate comprehensive work-resumption policies in advance. As an editorial in *The Lancet* stated, China’s experience will be crucial to understanding how to lift restrictions safely.¹¹ In addition, at a personal level, Chinese scholars have shared their own experiences with the world on how to cope with COVID-19. For example, some scholars shared how to face COVID-19 anxiety and encouraged scholars in other affected areas to keep an open mind about teaching reform necessary as part of the response;¹² other scholars shared their reflections on the prevention and containment measures and suggested cautious reporting of potential drugs.¹³

China’s biotechnology legal system has provided legal guidance for controlling the spread of COVID-19. The Law of Prevention and Treatment of Infectious Diseases (Infectious Diseases Prevention Law), promulgated in 1989 and revised after the SARS event¹⁴ in 2004, with further articles added in 2013, is the fundamental law for COVID-19 prevention and containment in China.¹⁵ Article 3 and Article 4 of the Infectious Diseases Prevention Law provides for a structured or tiered management of infectious diseases and divides them into Class A, Class B, and Class C;¹⁶ it also stipulates the specific responsibilities of each epidemic prevention organization in its Article 5, Article 6, and Article 7, i.e., how governments at various levels shall direct the work of preventing and treating infectious diseases. The health administrative department under the State Council shall be in charge of the work of preventing and treating infectious diseases, as well as exercising supervision and control over such diseases nationwide; and disease prevention and control institutions at all levels shall do the work of monitoring and forecasting infectious diseases, of conducting epidemiological investigation and reporting, as well as performing the work of preventing and controlling other diseases. The Infectious Diseases Prevention Law also contains essential chapters on Prevention of Infectious Diseases (chapter 2), Reporting on, Releasing Information on, and Announcing the Epidemic Situation (chapter 3), and Control of the Epidemic Situation (chapter 4).

On January 20, 2020, pursuant to the Infectious Diseases Prevention Law and approved by the State Council, the National Health Committee recognized COVID-19 as a Class B infectious disease but took the measures for prevention and contain-

ment appropriate under Class A. Based on this regulation, the State Council, governments at all levels, and medical agencies will be further empowered by law to take more active pandemic prevention measures to contain COVID-19 transmission. For example, Wuhan Command Centre of Novel Coronavirus Pneumonia Prevention and Control (CCNCPPC¹⁷), following the Infectious Diseases Prevention Law and other laws, announced the shutdown of the city and closed the exits from Wuhan, effectively curbing the spread of COVID-19.

Another biotechnology law that applies in conjunction with the Infectious Diseases Prevention Law is the Vaccine Administration Law. On December 1, 2019, China implemented its first vaccine administration law. Before the Vaccine Administration Law, China had been relying on the Regulation on the Administration of Circulation and Vaccination of Vaccines (Vaccine Administration Regulation) and Pharmaceutical Administration Law to regulate the circulation and administration of vaccines. However, the Vaccine Administration Regulation is only a government regulation rather than law, and the Pharmaceutical Administration Law focuses on drug safety and only sets general regulations for drug administration. This regulation and legislation fail to integrate the whole vaccine

¹¹COVID-19: *Learning from Experience*, 395(10229) THE LANCET (Editorial) 1011 (Mar. 28, 2020), [https://doi.org/10.1016/S0140-6736\(20\)30686-3](https://doi.org/10.1016/S0140-6736(20)30686-3)

¹²Kai Liu, *How I Faced My Coronavirus Anxiety*, 367(6484) SCIENCE 1398 (Mar. 20, 2020).

¹³Yonghong Xiao, *Taking Right Measures to Control COVID-19*, 20 THE LANCET 523 (Mar. 5, 2020).

¹⁴The 2002–2004 SARS outbreak was an epidemic involving severe acute respiratory syndrome (SARS) caused by severe acute respiratory syndrome coronavirus. The outbreak was first identified in Foshan, Guangdong, China, on November 16, 2002. Over 8,000 people from 29 different countries and territories were infected, and at least 774 died worldwide. SARS directly promoted the amendment of the Infectious Diseases Prevention Law (1989 version), and the revised Infectious Diseases Prevention Law (2014 version) enhanced China’s capacity for early warning and emergency disposal of infectious disease.

¹⁵Cao Yanlin and Chenguang Wang, *Discussion on Improving the Legal Regulation of Epidemic Prevention and Control in China*, 3 CHINA HEALTH LAW 57, 57 (May 2020).

¹⁶Infectious Diseases Prevention Law classifies infectious diseases into Class A, Class B, and Class C and enumerates all the common infectious diseases in an enumerative legislation mode. Infectious Atypical Pneumonia is Class B.

¹⁷CCNCPPC is short for Command Centre of Novel Coronavirus Pneumonia Prevention and Control, which is responsible for specific COVID-19 prevention and containment work.

chain of development, production, circulation, monitoring, handling abnormal reactions, safety measures, supervision and administration, and legal liability.¹⁸ The Vaccine Administration Law focuses on the whole vaccine chain and establishes the regulatory framework for vaccine administration, which provides legal guidance for orderly production of vaccines.¹⁹ Article 3 of the Vaccine Administration Regulation says “the state shall implement the most stringent management rules for vaccines,” but it also introduces the concept of risk management, which is reflected in Article 20 and Article 29,

for vaccines urgently needed to respond to major emergent public health incidents or other vaccines that are urgently needed as determined by the competent health department of the State Council, if benefits outweigh risks upon assessment, the medical products administration of the State Council may conditionally approve vaccine registration applications. In the case of any particularly serious public health emergency or any other serious emergency that poses a serious threat to public health, the competent health department of the State Council shall offer a proposal on the urgent use of the vaccine as required for the prevention and control of the infectious disease, and the vaccine may be put into urgent use within a certain scope and period with the approval upon organized demonstration by the medical products administration of the State Council.

Furthermore, “Vaccine lot release inspection items and frequency shall be dynamically adjusted according to the assessment of vaccine quality risks.” These risk management articles show that the Vaccine Administration Law is based on the understanding there is no “risk-free” vaccine and that a risk-benefit assessment should be applied to vaccines. Based on such legal guidance, China officially launched urgent use of three COVID-19 inactivated vaccines on July 22, 2020,²⁰ two from China National Biotec Group and one from Sinovac Biotech. These vaccines perform satisfactorily at present, with hundreds of thousands of people vaccinated without a single adverse reaction or infection.

The emergency legal system provides legal guidance for the emergency response to pandemic prevention and containment. China’s emergency legal system is relatively sound and has established relevant legal rules. After major disaster events, such as the catastrophic flood in 1998, the SARS outbreak

in 2003, or the massive Wenchuan earthquake in 2008, China’s emergency legal system has been serially improved, drawing on the lessons of each event. At present, there are more than ten emergency laws and regulations in China, enabling a fast connection between the central government’s decision(s) and the local governments’ implementation.²¹ Moreover, the government’s fast emergency response is critical for coping with major infectious diseases, such as hepatitis, tuberculosis, AIDS, influenza, hand-foot-and-mouth disease, dengue, and measles, as well as epidemics such as schistosomiasis, malaria, hydatid disease, and COVID-19. The Regulation on Responses to Public Health Emergencies (Public Health Emergencies Regulation) provides in Article 3 that “when an emergency outbreaks, the State Council shall set up a headquarters for the urgent handling of the national emergency, which shall be composed of the relevant departments of the State Council and the relevant departments of the army.” Article 4 states that “when an emergency breaks out, the government of the provinces, autonomous regions, or municipalities directly under the Central Government shall set up local headquarters for the urgent handling of the emergency.” These two articles provide legality protection for the establishment of national and provincial CCNCPPC, which can quickly establish a unified emergency response command system and give full play to the advantages of the nationwide system of “concentrating resources to accomplish large undertakings.”

The Emergency Response Law stipulates in Article 3 that “the natural disasters, accidental disasters and public health incidents shall be divided into four levels: especially serious, serious, large, and ordinary,” ensuring that provincial governments legally initiate first-level response measures to take the most extensive and strongest emergency prevention and containment measures such as closing

¹⁸Song Hualin, *System Construction and Legal System Innovation of Vaccine Management: Legislative Interpretation of The Vaccine Administration Law*, 7 CHINA FOOD & DRUG ADMIN. MAG. 4, 5 (July 15, 2019).

¹⁹*Id.* at 6.

²⁰In addition, China’s military had also granted approval to use the Ad5-nCOV vaccine from CanSino Biologics on June 25, 2020. However, this vaccine is currently only approved within the military for one year, and can’t be expanded without the approval of the Logistics Support Department of the Central Military Commission.

²¹Guo Feng, *Discussion on Constructing the Legal System of Epidemic Prevention and Control in China*, 3 LAW AND ECONOMY 3, 4 (May 15, 2020).

communities to travel, or ensuring that the Thunder God Mountain hospital, Fire God Mountain hospital, and other emergency hospitals were completed within 20 days.²² The Frontier Health and Quarantine Law provides in Article 16 that “Frontier Health and Quarantine Offices shall be authorized to require persons on entry or exit to complete a health declaration form and produce certificates of vaccination against certain infectious diseases, a health certificate or other relevant documents,” while Article 17 authorizes the Frontier Health and Quarantine Offices to “issue medical convenience cards, keep inspection or take other preventive or control measures, for persons who suffer from infectious diseases to be monitored, who come from areas in foreign countries where infectious diseases to be monitored are epidemic or who have close contact with patients suffering from infectious diseases to be monitored.” After the outbreak of COVID-19, the Chinese Customs agency, in accordance with the Frontier Health and Quarantine Law, adopted centralized quarantine and other medical observation measures for those who had epidemic area sojourn history. The local spread of COVID-19 in China has basically been interrupted, while the Frontier Health and Quarantine Law provides strong legal protection against importing cases from abroad.

The animal protection legal system provides legal support for eradicating COVID-19 from the biologic source. Scientists around the world have found that the COVID-19 originated from wild animals—most likely from Yunnan Bats.²³ The probable cause of the outbreak of COVID-19 is indiscriminate eating of wild animals. Therefore, the immediate rectification of the wild animal market can help prevent people from coming in contact with infectious viruses hidden in high-risk animals, which is of great significance for the eradication of COVID-19 from its biological source. Therefore, the Wild Animal Protection Law, which had once received little attention in China, immediately became crucial for eradicating the virus from the biological source. The Wild Animal Protection Law has gone through four amendments. The law’s general thrust is to protect key wildlife and their habitats from the perspectives of “animal welfare” and “rational utilization,” while only Article 30 states that “it shall be prohibited to illegally purchase wild animals under state priority conservation and their products for eating.” Based on Article 30, the Standing Committee of the national and local people’s congress have issued urgent regulations to control the wildlife market and essentially exclude eating from the “rational utilization” of wild-

life resources.²⁴ For example, the National People’s Congress Standing Committee Council approved a total ban on eating and illegally trading wild animals,²⁵ and the Guangdong Provincial People’s Congress Standing Committee also prohibited the eating and trading of wild animals.²⁶ All these committees wanted to eradicate viral transmission from the biological source.

In addition, the Animal Epidemic Prevention Law provides in Article 31 and Article 32 that “the government at or above the county level shall promptly take compulsory measures such as blocking, isolating, catching and killing, exterminating, sterilizing, harmlessly treating or giving emergency immunization to disease-infected animals for swiftly keeping the epidemic under control and wiping it out.” For example, after the outbreak of COVID-19 in Wuhan South China Seafood Market, the local government immediately sealed off the market and sterilized it, as part of the effort to eradicate the virus from the biological source.

3. PROBLEMS IN CHINA’S BIOTECHNOLOGY LAW DURING PANDEMIC PREVENTION AND CONTAINMENT

Every major catastrophe of mankind is a test of the country’s institutions and governance capacity;

²²In general hospital construction is governed by Land Administration Law, Urban and Rural Planning Law, Construction Law, and so on, and there are a number of legal approval procedures which need to be completed, *e.g.*, project approval, planning, land acquisition, bidding, construction, completion inspection, etc. However, Article 49 of the Emergency Response Law authorizes the government to build emergency facilities (like emergency hospitals) expeditiously.

²³Kristian G. Andersen, Andrew Rambaut, W. Ian Lipkin, Edward C. Holmes, and Robert F. Garry, *The Proximal Origin of SARS-CoV-2*, 26 NATURE MED. 450 (Mar. 17, 2020); Peng Zhou, Xing-Lou Yang, Zheng-Li Shi, *et al.*, *A Pneumonia Outbreak Associated with a New Coronavirus of Probable Bat Origin*, 579 NATURE 270 (Feb. 3, 2020).

²⁴The Standing Committee of the National People’s Congress is the legislature in China.

²⁵See the Decision of the National People’s Congress Standing Committee on Comprehensively Banning the Illegal Wildlife Trade, Getting Rid of the Habit of Eating Wildlife, and Ensuring the Life, Health and Safety of the People.

²⁶See the Decision of Guangdong Provincial People’s Congress Standing Committee on the Approval on Prevention and Control of COVID-19.

in face of every major catastrophe, the strengths and weaknesses of human nature and social institution will be fully exposed.²⁷ COVID-19 has not only tested the national governance capacity, but also exposed weaknesses in China's biotechnology legal system, including the emergency legal system and animal protection legal system.

Firstly, there was a lag in the early warning mechanism of the pandemic situation in the biotechnology legal system. According to a retrospective case analysis paper from the *New England Journal of Medicine*, there is evidence that human-to-human transmission had occurred among close contacts since the middle of December 2019.²⁸ It was not until January 20, 2020, however, that human-to-human transmission had been confirmed by Dr. Zhong Nanshan in the CCTV interview. Some government officials admitted that the disclosure of the pandemic information was not timely, because the local government must be authorized, according to the law, to disclose the novel coronavirus pandemic. The late warning shows that there are weaknesses or gaps in the current pandemic warning system of China's biotechnology legal system. Specifically, first of all, the responsible organs for early warning of the pandemic situation are not unified. According to Article 19 of the Infectious Diseases Prevention Law, early warning of infectious diseases should be issued by the health administrative department under the State Council and the provincial government in time; while according to Article 43 of the Emergency Response Law, as the COVID-19 is a public health event, local and central government should issue warnings and announce that they are entering an early warning period; at the same time, the National Emergency Response Plan for Public Health Emergencies also stipulates that public health event warnings should be issued by the government health administrative departments of all levels.²⁹

Secondly, there are drawbacks in the mechanism coping with "unexplained infectious disease." Measures for the prevention and containment of unexplained infectious diseases should be promulgated to the public only by the health administrative department under the State Council, with the approval of the State Council; *i.e.*, the "unexplained infectious disease" found in a locality needs to be reported to the State Department. In theory, Infectious Diseases Prevention Law cannot be applied to prevent and contain the "unexplained infectious disease" before it is confirmed by the State Council; however, the virus may have caused disastrous consequences before that occurs. Besides, the Diseases

Prevention and Control Institutions (DPCI³⁰) in China enjoy a marginal legal status. Article 78 of the Infectious Diseases Prevention Law defines the DPCI as "the disease prevention and control centers engaged in disease prevention and control as well as the units engaged in professional activities similar to those of the said institutions," *i.e.*, the government-sponsored institution undertaking public functions, not the administrative agency. The activities of DPCI, therefore, belong to the intermediate link in the administrative process and undertake the functions of consulting and supporting. DPCI makes recommendations to the health administrative departments and helps them make decisions with external legal effects.³¹ As a result, the highly specialized DPCI, with the most abundant frontline information on a pandemic situation, has no right to issue an early warning. Its legal status makes it lack enough professional autonomy and independence to make full use of the information it gathers.³²

Thirdly, the emergency legal system lacks a comprehensive top-level design. The responsible organ for initiating an emergency plan is inconsistent. Article 44 of the Emergency Response Law authorizes governments above the county level to initiate emergency plans, but Article 20 of Infectious

²⁷Yu Keping, *The Impact of the COVID-19 Crisis on National Governance*, PKU RESEARCH CENTER FOR CHINESE POLITICS (Apr. 30, 2020), <http://www.chinagovernance.pku.edu.cn/mzyt/112413.htm>

²⁸Qun Li, M.Med., Xuhua Guan, *et al.*, *Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus-Infected Pneumonia*, 382 NEW ENGLAND J. MED. 1199, 1207 (Mar. 26, 2020).

²⁹Although the National Emergency Response Plan for Public Health Emergencies belongs to the normative documents of the State Council and is far lower than the first two laws in effect, it is issued by China's highest administrative organ: as such, the document is binding on all governments throughout the country and the direct basis for each local emergency response plan to be formulated. See Jin Ziling, *The Early Warning System of Public Health Emergencies from the Perspective of Risk*, 3 CONTEMP. L. REV. (June 2020).

³⁰DPCI is short for Diseases Prevention and Control Institutions, which plays the same role as the Centers for Disease Control and Prevention.

³¹Song Hualin, *Reflection and Reconstruction of the Legal Status of Diseases Prevention and Control Institutions*, 4 EXPLORATION AND FREE VIEWS 195 (Apr. 20, 2020).

³²Gu Xin, *The Power of Knowledge and the Introduction of Social Governance: The Improvement of the Early Warning System of the Outbreak*, 2 GOVERNANCE STUD. 32 (Feb. 18, 2020).

Diseases Prevention Law requires governments above the county level to initiate the emergency plan only after receiving warning of infectious diseases from the State Council or a provincial health administrative department. It can be concluded that the Emergency Response Law lays more emphasis on territorial management and imposes emergency responsibilities on the local government, while the Infectious Diseases Prevention Law emphasizes the leading role of health administrative departments in pandemic response.³³ Moreover, the neglect of local self-government organizations leads to a lack of legal basis for some emergency measures. For example, residents' and villagers' committees have taken measures during the pandemic to contain it, such as sealing the doors of potentially infected persons and blocking or even destroying roads. It is of great importance for local self-government organizations to actively respond to and participate in pandemic prevention and containment; it is not only effective but also legal to adopt many containment measures to cut off the COVID-19 transmission route.³⁴ However, it is possible to go too far, and in practice, a lot of extreme and illegal actions, such as locking the private doors of Wuhan returnees and blocking roads without authorization, were taken. This is because the current emergency legal system pays too much attention to the government and health administrative departments, but ignores the power of local self-government organizations.³⁵ This leads to a lack of detailed rules for the implementation of relevant measures and a fuzzy boundary of power.

Furthermore, some of the pandemic prevention and containment policies changed frequently—even daily (or more frequently than daily). For example, Wuhan CCNPPC issued the Notice on Strengthening the Management of Vehicles and Personnel Entering and Leaving Wuhan (No.17 Notice) on February 24, 2020, allowing people to leave the city in staggered shifts, but then issued the No. 18 Notice to declare the No. 17 Notice invalid only three hours later. The No. 17 Notice was issued by Wuhan CCNPPC publicly with legal effect, but was then declared invalid by a superseding announcement without formally revoking it, which was procedurally improper. Similarly, Shantou, in Guangdong Province, the CCNPPC issued a “Lock-down Notice” on March 11, 2020, which was effectively withdrawn by a superseding announcement; a similar thing also happened with Qianjiang CCNPPC, in Hubei Province, on their work resumption notice. The specific pandemic situation is unpredictable, so the notices from CCNPPC

may indeed be wrong and an improper announcement should be revoked. However, the improper announcement should be revoked by the appropriate subject through a proper withdrawal procedure, rather than by simply posting a new, contradictory notice, which can lead to confusion. A rational withdrawal mechanism for official notices also needs to be considered in the top-level design of the emergency legal system.

Finally, the adjustment scope of the animal protection legal system is too narrow. The current Wild Animal Protection Law regulates only “the rare and endangered species of terrestrial and aquatic wild animals and the terrestrial wild animals of significant ecological, scientific, or social value,” but the animals that may cause anthroozoonoses include not only the aforesaid “valuable animals” but also “non-valuable animals,” and many animals that are not protected under the Wild Animal Protection Law are still at risk of spreading the virus. For example, pigeons may cause H7N9 bird flu, the paguma larvata may carry SARS virus, fruit bats may carry the Ebola virus, the Aedes is linked to Zika virus, and the Yunnan bat may carry novel coronavirus. None of these animals are in the List of Wild Animals Under State Priority Conservation, and the current Wild Animal Protection Law fails to prohibit people from eating these high-risk wild animals. Thus, the current Wild Animal Protection Law cannot satisfy the legal need to eradicate the viral spread from the biological source. While the Chinese nation has a splendid food culture, the consumption of these wild animals has

³³Lin Hongchao, *To Test and Improve the Legal System of Emergency Response During the Pandemic*, 2 CHINA EMERGENCY MGMT. 24 (Feb. 15, 2020).

³⁴Both the Infectious Diseases Prevention Law and Emergency Response Law authorize governments at or above the county level to restrict movement, while Article 9 of the Infectious Diseases Prevention Law and Article 40 of the Public Health Emergencies Regulation also make it clear that local self-government organizations have the responsibility and obligation to assist governments at or above the county level take pandemic prevention and containment measures. See Zheng Lin, *Legal Thoughts on Measures to Close Doors and Roads during the Pandemic Prevention and Contain Period*, 2(178) PUBLIC SECURITY SCIENCE J., 61-66, 62 (April 20, 2020).

³⁵The Infectious Diseases Prevention Law and the Public Health Emergencies Regulation each contain only one provision on emergency measures for local self-government organizations; and there are only three such provisions in the Emergency Response Law.

increased the possibility of human contact and infection with the virus, which creates huge public health risks.

4. AMENDMENT OF CHINA'S BIOTECHNOLOGY LAWS UNDER PANDEMIC PREVENTION AND CONTAINMENT

To deal with the problems exposed during the pandemic prevention and containment process, the whole country unanimously requested to amend the current biotechnology legal system to make up for the shortcomings. On February 5, 2020, President Xi Jinping made a special request to improve relevant legislation on pandemic prevention and containment; the speaker of the third session of the 13th National People's Congress (NPC)³⁶ expressed that the Standing Committee of the NPC plans to enact and amend 17 laws this year and in the coming year, and modify 13 other laws, to order to improve and strengthen the biotechnology legal system; the spokesman for Legal Working Committee of the NPC also expressed that they have already deployed work on amending the Wild Animal Protection Law, and planned to add the amendment of the Wild Animal Protection Law to the Standing Committee's legislative work plan this year. They will also speed up the process of amending the Animal Epidemic Prevention Law and other laws; during the 13th NPC in May this year, deputies of the NPC have also put forward pragmatic and feasible proposals for amending the laws, which not only urges China to improve the biotechnology legal system of pandemic prevention and containment, but also contributes Chinese solutions to the global pandemic prevention and containment.

4.1. *Comprehensively and systematically amend the biotechnology legal system for pandemic prevention and containment*

At the third session of the 13th NPC, the deputies proposed a package of amendments to the relevant laws and regulations on pandemic prevention and containment. The specific proposals should be led by the relevant special NPC committees, with coordinating assistant from Supreme People's Court, Supreme People's Procuratorate, and relevant functional departments of the State Council, making overall arrangements and promptly amending relevant laws on pandemic prevention and containment.

At present, China's legal systems for pandemic prevention and containment include the biotechnology legal system, the emergency legal system, and the animal protection legal system. There are problems such as insufficient connection and even contradictions both within each legal system and between the legal systems. Some deputies pointed out that there were conflicts between the Infectious Diseases Prevention Law and the Emergency Response Law, and the application of the law was not uniform; some deputies also suggested that the Infectious Diseases Prevention Law and the Emergency Response Law should be amended in a coordinated and synchronized manner; some deputies also suggested strengthening the connection among agriculture and forestry departments, public health departments and DPCI, so as to improve the joint prevention, control, and management system for anthroponozoonoses.

At the same time, some laws are also outdated. For example, the Public Health Emergency Regulations were amended in 2011, the Infectious Diseases Prevention Law was amended in 2013, and the Animal Epidemic Prevention Law was amended in 2015. As globalization continues to deepen and public health risks are increasing, it is necessary to update relevant biotechnology laws promptly. Only by strengthening top-level design and comprehensively designing a set of cohesive and complementary legal systems for pandemic prevention and containment, and bringing newly emerging infectious diseases with high transmissibility and mortality into the scope of statutory infectious diseases, can we respond more quickly and protect the lives and health of the people.

4.2. *A total ban on eating terrestrial wild animals*

Indiscriminate eating of wild animals has seriously threatened public health, and all parts of society have expressed strong objection to this. In order to control and eradicate the novel coronavirus from the biologic source, to make up for the shortcomings of Wild Animal Protection Law, the sixteenth meeting of the 13th NPC Standing Committee on February 24, 2020, issued the Decision on Comprehensively Banning the Illegal Wild Animal Trade, Abolishing the Habit of Indiscriminate Consumption of Wild Animals, and Ensuring the Life, Health, and Safety of the People (Wild Animal Protection Decision), which went into effect on the same day. The Wild

³⁶NPC is short for National People's Congress.

Animal Protection Decision states in Article 2 that “the consumption of terrestrial wild animals of important ecological, scientific, and social value and other terrestrial wild animals protected by the state, including terrestrial wild animals bred and raised in captivity, shall be comprehensively prohibited.” This brand-new set of regulations changed the “limited fasting principle” adhered to by the Wild Animals Protection Law into a “comprehensive fasting principle”:³⁷ that is, as long as the terrestrial wild animals grow and breed naturally in the wild environment, no matter whether they are under special protection or whether they are bred artificially, they must not be eaten as food.

As for aquatic wild animals, they are free to be eaten. This is because it is very difficult for aquatic wild animals to be infected with the coronavirus: the important route of transmission of coronavirus is the spread of respiratory droplets, while many aquatic wild animals, such as fish, have no lungs, so there is no physiological basis for droplet transmission. An article published on bioRxiv on May 1, 2020, also suggested that SARS-CoV-2, *i.e.*, the novel coronavirus, can infect a broad range of mammals, but few fish, birds, or reptiles.³⁸ Therefore, there is currently no need for a total ban on eating aquatic wild animals.

The specific basis for the current implementation of the “comprehensive fasting principle” is the National Catalogue of Livestock Genetic Resources. Article 3 of Wild Animal Decision sets that animals listed in the National Catalogue of Livestock Genetic Resources belong to livestock and poultry, and the provisions of the Animal Husbandry Law shall apply. According to this article, as long as wild animals are listed in the National Catalogue of Livestock Genetic Resources, they will be regarded as livestock and poultry, which can be traded and eaten,³⁹ and the terrestrial animals that fail to be listed in the National Catalogue of Livestock Genetic Resources are wild and cannot be eaten. In line with the “comprehensive fasting principle,” Ministry of Agriculture and Rural Affairs officially announced the National Catalogue of Livestock Genetic Resources on May 29, 2020, and defined for the first time 33 kinds of livestock and poultry, which further clarified what kinds of terrestrial animals can be eaten and what kinds cannot. Although the Wild Animals Protection Law has not been amended, it will certainly absorb the relevant spirit and provisions of the Wild Animals Protection Decision, continue to adhere to the “comprehensive fasting principle” of terrestrial wildlife, and intend to prevent people from coming in contact with high-risk animals po-

tentially harboring infectious viruses, with a purpose of eradicating the anthroponozoonoses from biological sources.

4.3. *The future laws will focus on enhancing the early warning capabilities of infectious disease prevention and containment*

“Better safe than sorry”: the biotechnology legal system has long been more prone to damage prevention; *i.e.*, prevention attracts more attention than precaution.⁴⁰ But if precaution can be well-structured, it is possible to avoid adverse consequences in the future and minimize the losses. One of the forms of communication of risk information is early warning: enhancing the early warning capability of infectious disease prevention and containment may therefore become the focus of the amendment. In response to this problem, some NPC deputies suggested that the early warning authority of the Infectious Diseases Prevention Law should be amended, and the authority for issuing an early warning of infectious diseases enjoyed by the health administrative department under the State Council and provincial governments should be delegated to the provincial health administrative departments—*i.e.*, the provincial health administrative departments should have the authority to issue prompt early warnings to reduce delays due to administrative approval. Other deputies suggested simplifying the pandemic reporting procedure, establishing a direct connection for pandemic reporting between the pandemic area government and the central competent authority, and empowering medical staff in the pandemic area to make the early warning advice.

³⁷The Wild Animals Protection Law upholds “limited fasting principle,” because it only penalizes the conduct of production, sale, and purchase, not consumption, and the animals protected must be those under the State’s key protection. See Ye Liangfang, *Discussion on the Application of Wild Animals Protection Decision*, 3 RES. ON RULE OF LAW 24, 31 (May 7, 2020).

³⁸S.D. Lam, N. Bordin, V.P. Waman, *et al.*, *SARS-CoV-2 Spike Protein Predicted to Form Complexes with Host Receptor Protein Orthologues from a Broad Range of Mammals*, bioRxiv (May 1, 2020), doi: <https://doi.org/10.1101/2020.05.01.072371>

³⁹Ye Liangfang, *Discussion on the Application of Wild Animals Protection Decision*, 3 RES. ON RULE OF LAW 24, 34 (May 7, 2020).

⁴⁰Jin Ziling, *The Early Warning System of Public Health Emergencies from the Perspective of Risk*, 3 CONTEMP. L. REV. (June 2020).

4.4. The future laws will expand the applicable scope of Infectious Diseases Prevention Law

It was pointed out that the Infectious Diseases Prevention Law lacks provisions on “unexplained infectious disease” and, in particular, the Infectious Diseases Prevention Law cannot be applied until the virus strain and infectivity have been confirmed and reported to the health administrative department under the State Council. To address this problem, some deputies believed that the applicable scope of the Infectious Diseases Prevention Law

should be expanded. Therefore, the applicable scope of the Infectious Diseases Prevention Law is expected to be expanded in the following amendment process.

The amendments to these biotechnology laws have laid a solid foundation for the prevention and containment of COVID-19 in China and also provided the world with the benefit of Chinese solutions.

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